Wigner Resolution from Action Capacity

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The Wigner function represents a quantum state in phase space. Smoothing it gives a non-negative portrait, with the smoothing scale conventionally left free. We show that the scale is set by the state. The state's action capacity $\pi\,\Delta x\,\Delta p$ fixes a family of de Gosson quantum blobs deforming at constant action $h/2$. Convolving the Wigner function with the kernel of a single $h/2$ blob renders it non-negative, one coordinate at a time. Integrating over the family is coordinate-free and resolves the portrait at $\pi\hbar^2/(\Delta x\,\Delta p)$, the symplectic polar dual of the capacity. The portrait is everywhere non-negative and holds the structure of the Wigner function in place. Wigner resolution derives from the state's action capacity alone and sharpens as that capacity grows.

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The Wigner function represents a quantum state in phase space [1, 2]. It is defined pointwise below Planck's quantum of action, where Hudson's theorem forces it negative [3]. Its negativity is a resource for quantum advantage [4, 5] and a measure of nonclassicality [6]. A non-negative distribution supports classical sampling [7–11] and follows from smoothing the Wigner function at a freely chosen scale. Here the state sets that scale without a free parameter.

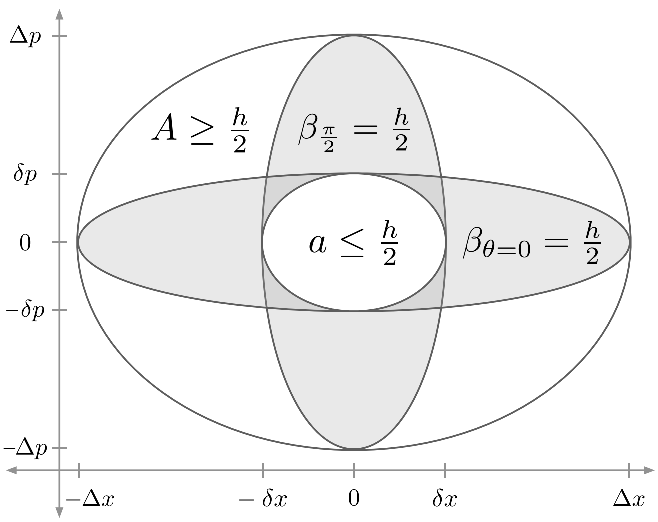

FIG. 1. **Capacity, quantum blobs at principal axes, and resolution.** The state's covariance ellipsoid has action capacity $A = \pi\,\Delta x\,\Delta p \geq h/2$. The quantum blob family (Fig. 2) has two members at the principal axes, $\beta_{\theta=0} = \pi\,\Delta x\,\delta p = h/2$ and $\beta_{\pi/2} = \pi\,\delta x\,\Delta p = h/2$. Zurek's relations $\delta p \sim \hbar/\Delta x$ and $\delta x \sim \hbar/\Delta p$ locate the semi-axis scales from the state's action capacity. The resolution a is an ellipse with semi-axes $\delta x, \delta p$ and action $a = \pi\,\delta x\,\delta p = \pi\hbar^2/(\Delta x\,\Delta p)$. A circumscribes $\beta_{\theta=0}$ and $\beta_{\pi/2}$, which circumscribe a.

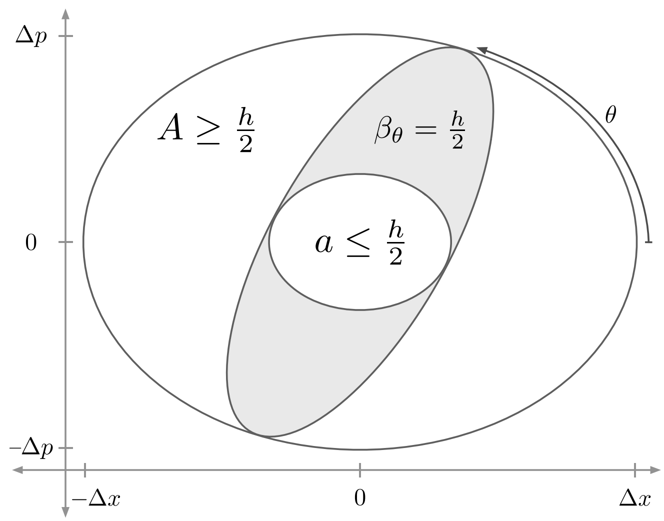

FIG. 2. **Capacity, quadrature family of quantum blobs, and resolution.** Action capacity A circumscribes the quantum blob family $\{\beta_\theta\}$, which circumscribes resolution a. Across θ the blob family deforms at constant action $\beta_\theta = h/2$. At the principal angles $\theta = 0$ and $\theta = \pi/2$, the family recovers $\beta_{\theta=0}$ and $\beta_{\pi/2}$ from Fig. 1.

Planck divided phase space into elementary regions of action [12]. Heisenberg's uncertainty principle $\sigma_x\,\sigma_p \geq \hbar/2$ [13–15] fixes their minimum area but not their form, which a rotation of the axes redraws [16]. Geometrically, de Gosson replaced the coordinate-dependent cell with the state's covariance ellipse of semi-axes $\Delta x = \sqrt{2}\,\sigma_x$ and $\Delta p = \sqrt{2}\,\sigma_p$. Its symplectic capacity $\pi\,\Delta x\,\Delta p$ is a canonical invariant, the area unchanged by rotation, squeeze, or shear [16, 17]. This is the action capacity (Figs. 1, 2),

$$A = \pi\,\Delta x\,\Delta p \geq h/2. \tag{1}$$

The Wigner function carries structure at small scales. Zurek located those scales, $\delta p \sim \hbar/\Delta x$ and $\delta x \sim \hbar/\Delta p$, from the state's action capacity [18]. We identify them as the principal axes of a family of de Gosson quantum blobs (Fig. 1),

$$\beta_{\theta=0} = \pi\,\Delta x\,\delta p = h/2, \qquad \beta_{\pi/2} = \pi\,\delta x\,\Delta p = h/2. \tag{2}$$

The capacities $\Delta x, \Delta p$ run along the major axes, the resolutions $\delta x, \delta p$ along the minor.

Between these principal members the family $\{\beta_\theta\}$ deforms across $\theta \in [0, \pi)$ at constant action $h/2$ (Fig. 2).

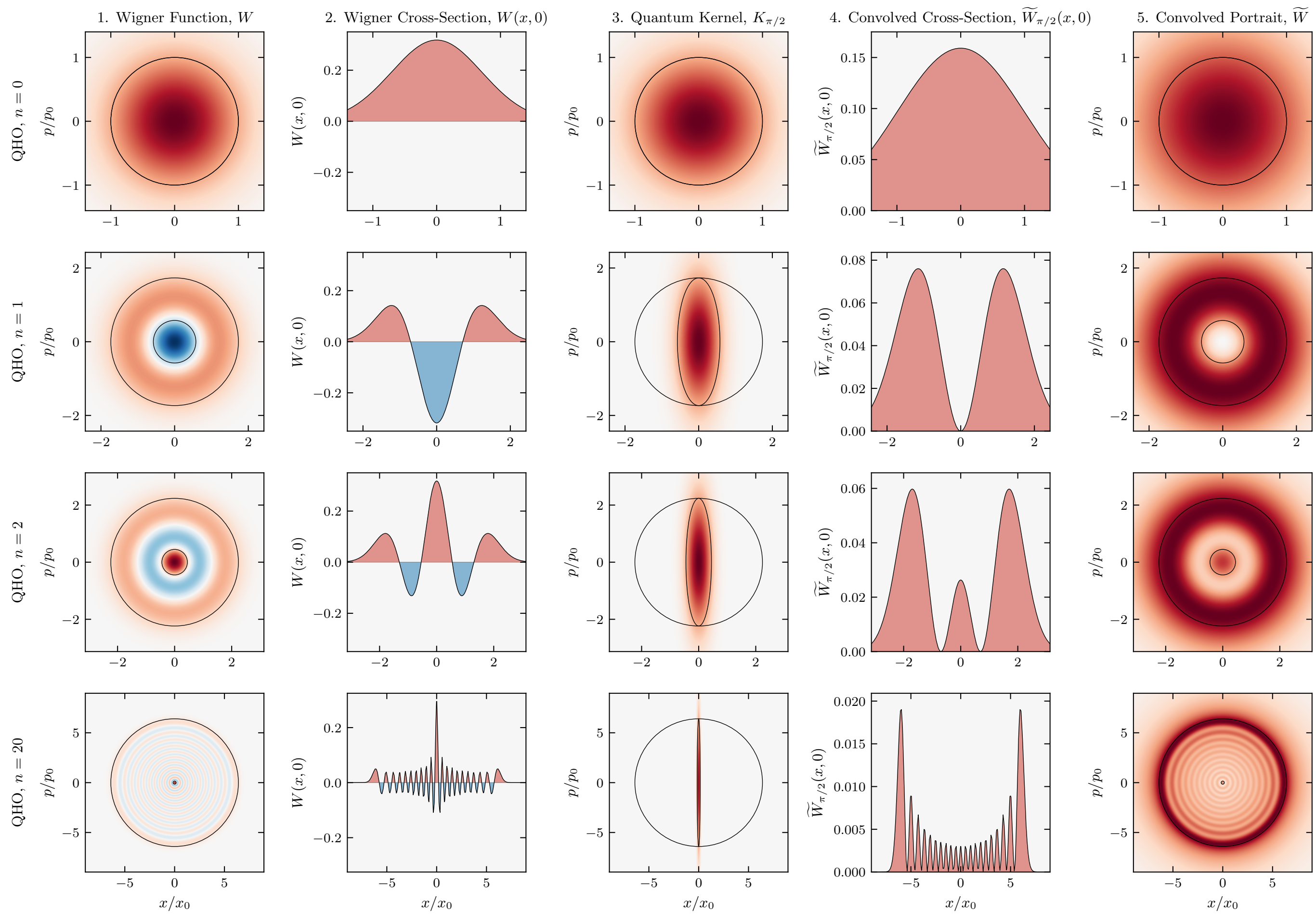

FIG. 3. **Harmonic oscillator.** Rows correspond to Fock states $n = 0, 1, 2, 20$ with action capacity $A/(h/2) = 2n+1$. Column 1: Wigner function $W(x,p)$ (negativity in blue) overlaid with the capacity A and resolution a. Centered at the covariance origin, a marks the resolution of the finest structure (the Zurek scale) of the bare, unconvolved Wigner function. Column 2: Cross-section $W(x,0)$ exhibiting negative dips. Column 3: Quantum kernel $K_{\pi/2}$ overlaid with A and the quantum blob $\beta_{\pi/2}$ (action $h/2$, restoring Planck's quantum of action). Column 4: Convolved cross-section $\widetilde{W}_{\pi/2}(x,0)$ at quadrature angle $\theta = \pi/2$, which is non-negative. Column 5: The portrait $\widetilde{W}(x,p)$, integrated over the blob family across all θ, overlaid with A and a (symplectic polar duals). It is everywhere non-negative, with positive lobes at the positions of those in W, resolved at a. For the vacuum ($n = 0$), A and a coincide as a Gaussian blob at $A = h/2$. As n increases, A grows and a reciprocally shrinks ($a\,A = (h/2)^2$). At $n = 20$, the rings condense toward a classical energy trajectory, leaving a as a fine central ellipse. Axes are in the oscillator scales $x_0 = \sqrt{\hbar/m\omega}$ and $p_0 = \sqrt{\hbar m\omega}$.

Capacity A circumscribes each blob β_θ. In the scaled coordinates $X = x/\Delta x$, $P = p/\Delta p$ every member is the same rigid ellipse rotated through θ,

$$\beta_\theta = \left\{(x,p) : X_\theta^2 + \tfrac{A^2}{(\pi\hbar)^2}\, P_\theta^2 \le 1\right\}, \tag{3}$$

inscribed in the unit disk. The quadratures $X_\theta = X\cos\theta + P\sin\theta$ and $P_\theta = -X\sin\theta + P\cos\theta$ are those of homodyne tomography at phase θ.

The Wigner function of the state is

$$W(x,p) = \frac{1}{\pi\hbar}\int \psi^*(x+s)\,\psi(x-s)\,e^{2ips/\hbar}\,ds. \tag{4}$$

The quantum kernel K_θ is a pure-state Wigner function, the centered Gaussian whose $h/2$ ellipse is the quantum blob β_θ,

$$K_\theta(x,p) = \frac{1}{\pi\hbar}\exp\left[-X_\theta^2 - \tfrac{A^2}{(\pi\hbar)^2}\, P_\theta^2\right]. \tag{5}$$

Convolving W with K_θ,

$$\widetilde{W}_\theta(x,p) = (W * K_\theta)(x,p), \tag{6}$$

lifts W to Planck's quantum of action and renders $\widetilde{W}_\theta$ everywhere non-negative [8, 9, 19, 20].

Zurek also located the area scale $a_Z \sim \hbar\times(\hbar/A)$ [18, 21] of the chessboard structure in the compass state (Fig. 5, column 1). We identify its coordinate-free form as the resolution a. It is the ellipse with semi-axes $\delta x, \delta p$

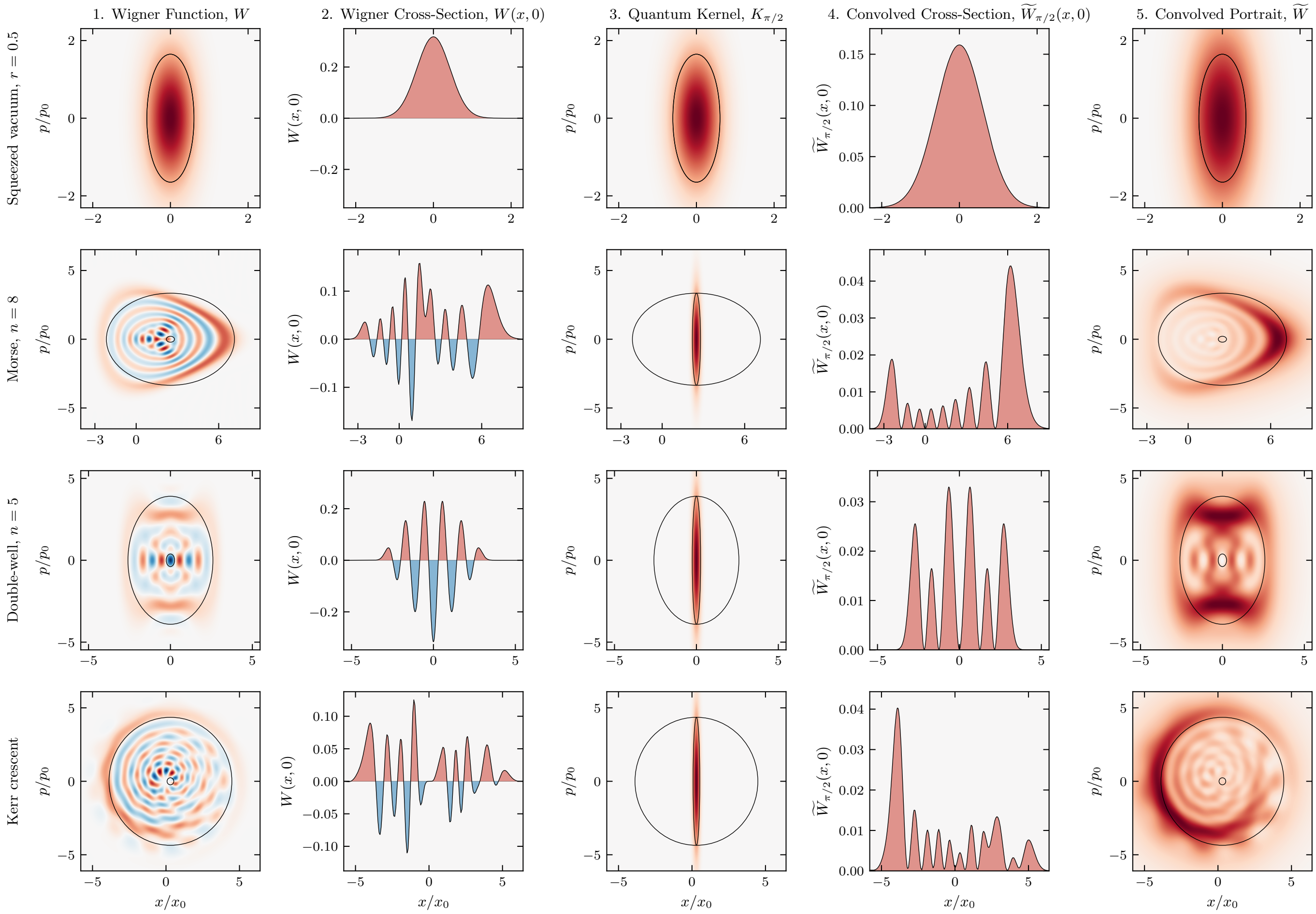

FIG. 4. **Squeezed vacuum, eigenstates, and the Kerr crescent.** Rows correspond to the squeezed vacuum ($r = 0.5$), the Morse eigenstate ($n = 8$), the double-well eigenstate ($n = 5$), and the Kerr crescent, a coherent state sheared at constant photon number. Columns are as in Fig. 3. The Kerr state is shown in the principal frame of its covariance, where the overlays are axis-aligned. The portrait $\widetilde{W}$ (Column 5) remains strictly non-negative across all four states, resolved at a.

(Figs. 1, 2) and action

$$a = \pi\, \delta x\, \delta p = \frac{\pi \hbar^2}{\Delta x\, \Delta p}. \tag{7}$$

It is the resolution of the state's finest structure in the bare Wigner function (Figs. 3, 4, 5, column 1) and the symplectic polar dual of the capacity A [22, 23]. The duality is reflexive, so A and a each fix the other, and antimonotone, so a shrinks as A grows. Each blob β_θ is circumscribed by A and circumscribes a, nesting as $A \supseteq \beta_\theta \supseteq a$ (Fig. 2). Resolution and capacity are reciprocal at the quantum of action, $a\, A = (h/2)^2$.

A single blob commits to a quadrature, a choice of coordinate the state does not favor. Integrating $\widetilde{W}_\theta$ over the family is coordinate-free, yielding the phase-space portrait,

$$\widetilde{W}(x,p) = \frac{1}{\pi} \int_0^\pi \widetilde{W}_\theta(x,p)\, d\theta. \tag{8}$$

Each $\widetilde{W}_\theta \geq 0$, so $\widetilde{W} \geq 0$ everywhere. Where standard tomography inverts the marginals to recover the Wigner quasi-probability W [24–26], convolving W with $\{K_\theta\}$ over the same θ builds the non-negative portrait $\widetilde{W}$.

The convolution and the integral are linear, so the family reduces to a single convolution,

$$\widetilde{W} = W * K, \qquad K = \frac{1}{\pi} \int_0^\pi K_\theta\, d\theta, \tag{9}$$

and the breadth of the kernel's Fourier transform $\widehat{K}$ sets the portrait's resolution. With (ξ_x, ξ_p) the chord conjugate to (x,p), integrating over θ bounds $\widehat{K}$ above by a single Gaussian,

$$\widehat{K}(\xi_x, \xi_p) \leq \exp\left[-\tfrac{1}{4}\left(\delta x^2\, \xi_x^2 + \delta p^2\, \xi_p^2\right)\right], \tag{10}$$

with equality at the origin. The bound cuts off at δx and δp, so the portrait carries no structure finer than $\pi\, \delta x\, \delta p = a$, derived exactly in the Supplemental Material [27].

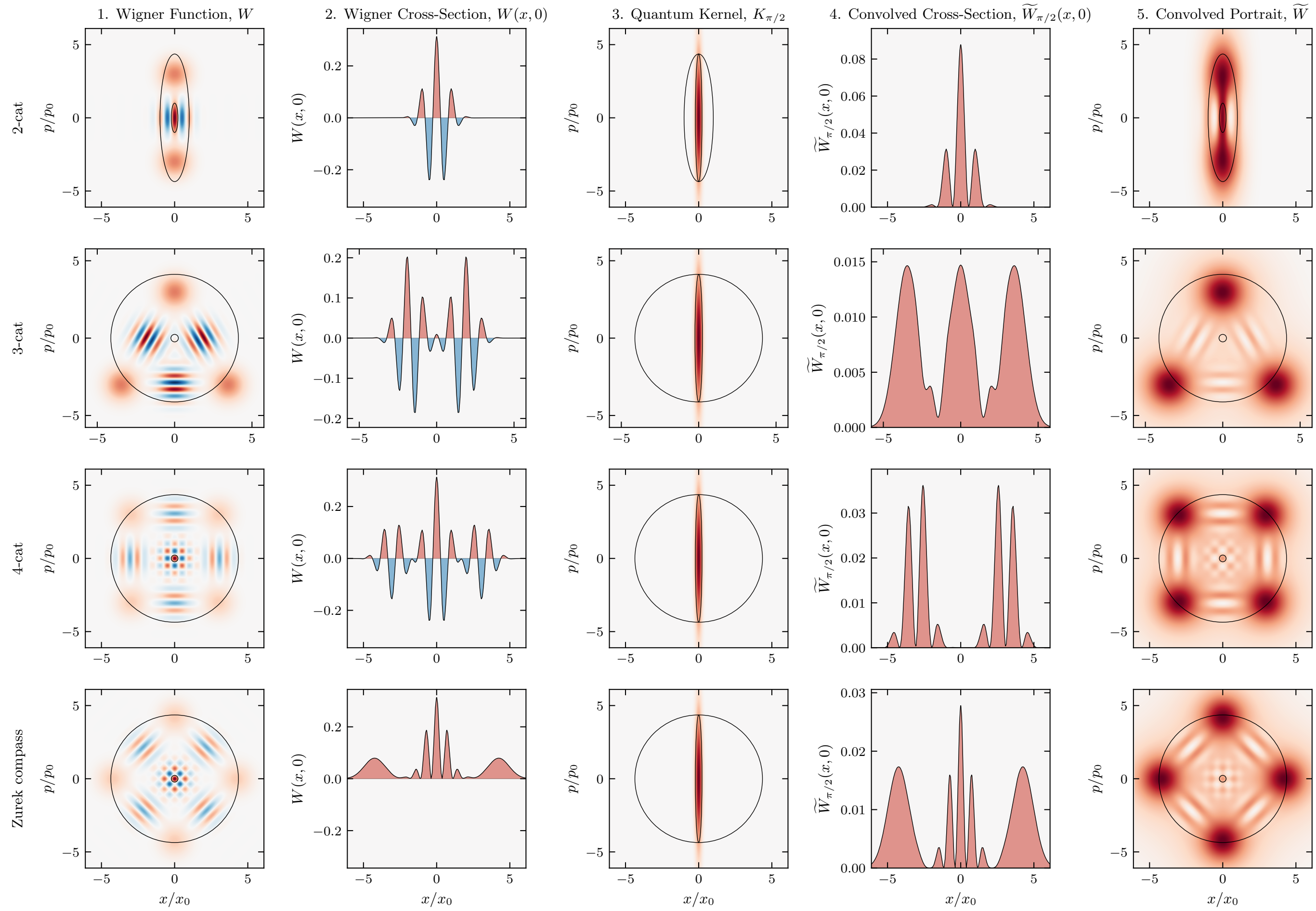

FIG. 5. **Cat states.** Rows correspond to the 2-cat, 3-cat, 4-cat, and Zurek compass states. Columns are as in Fig. 3. As before, the portrait $\widetilde{W}$ (Column 5) is everywhere non-negative, with positive lobes at the positions of those in W, resolved at a. The 4-cat and Zurek compass states differ by a $\pi/4$ rotation in phase space.

A quantum blob β_θ too large for A to circumscribe, or too small to circumscribe a, is a blob of another state and portrays that state, not this one [22, 28]. Every $h/2$ blob of this state is inscribed in A and circumscribes a, touching each, so no $h/2$ smoothing of this state resolves finer than a, and a kernel below $h/2$ fails to render the portrait non-negative [8, 9]. The finest resolution is a, reached by the family $\{\beta_\theta\}$ across $\theta \in [0, \pi)$.

The principal frame, where A is axis-aligned, costs no generality. The Wigner function transforms covariantly under the symplectic group [20], so the map that diagonalizes a state's covariance carries it to this frame. The Kerr crescent, a sheared coherent state, is resolved this way (Fig. 4). In one degree of freedom, A fixes the family uniquely, and the Zurek structure is already present for states with $A > h/2$. In higher dimensions, action $h/2$ no longer singles out a blob [16, 28], symplectic capacity and volume diverge, and the higher-dimensional case remains for future work.

A fixed vacuum-width Gaussian [7] smooths every state equally, matching only the vacuum $A = h/2$ and erasing the detail of larger states $A > h/2$. Methods with a free parameter [29–31] resolve more finely, the scale set from outside the state. Here the state sets its own scale. Its action capacity A fixes the resolution a as the symplectic polar dual, with nothing left free.

Figures 3, 4, and 5 show numerical results across twelve states. The portrait $\widetilde{W}$ is non-negative to machine precision, the chord-domain transfer carrying W to $\widetilde{W}$ follows Eq. (10), and the cross-spectral phase between W and $\widetilde{W}$ vanishes. Across every state the portrait $\widetilde{W}$ resolves W at a and displaces nothing [27].

The uncertainty product $\sigma_x \sigma_p$ grows unbounded, the micro-to-macro transition Heisenberg [32] and Bohr [33] identified as requiring care. The quantum blob does not. As A grows, the family $\{\beta_\theta\}$ deforms at fixed action $h/2$. Squeezing deforms a state by external apparatus [34]. Here the scale called sub-Planck [18] is the minor axis δ of an $h/2$ blob, narrowing as the conjugate capacity Δ grows. A Schrödinger cat with capacity $\Delta p \sim 1$ kg·m/s has resolution $\delta x = \hbar/\Delta p \sim 10^{-34}$ m, nineteen orders sub-nuclear [35]. The correspondence principle follows from $A \to \infty$ and $a \to 0$, the physical limit underlying the heuristic $h \to 0$. The resolution sharpens as the action capacity grows, $a = (h/2)^2 / A$. We read this structure as the geometry of phase space itself, which de Gosson finds underlying the uncertainty principle [17, 19, 22].

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